Question 4

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#include "ns3/core-module.h"
#include "ns3/network-module.h"
#include "ns3/internet-module.h"
#include "ns3/point-to-point-module.h"
#include "ns3/csma-module.h" // add the required library name
#include "ns3/applications-module.h"
#include "ns3/ipv4-global-routing-helper.h"

using namespace ns3;

NS_LOG_COMPONENT_DEFINE ("DCN_Lab_Final");

int main (int argc, char *argv[])
{
LogComponentEnable("UdpEchoClientApplication", LOG_LEVEL_INFO);
LogComponentEnable("UdpEchoServerApplication", LOG_LEVEL_INFO);

// Create nodes for LAN1 (n5, n6, n7, n0)

// Create nodes for LAN2 (n1, n2, n3, n4)
 NodeContainer p2pNodes;
 NodeContainer csmaNodes;  // right LAN1
 csmaNodes.Create (4);

 NodeContainer csmaNodes1; // left LAN2
 csmaNodes1.Create (4);
 p2pNodes.Add(csmaNodes.Get(3),csmaNodes1.Get(0));


InternetStackHelper stack;
stack.Install(csmaNodes); // fill the space
stack.Install(csmaNodes1 ); // fill the space
stack.Install(p2pNodes);

CsmaHelper csma;
csma.SetChannelAttribute("DataRate", StringValue("6Mbps")); // add the value
csma.SetChannelAttribute("Delay", TimeValue(MilliSeconds(2)));

NetDeviceContainer lan3Devices = csma.Install(csmaNodes1); // fill the space
```

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NetDeviceContainer lan2Devices = csma.Install(csmaNodes); // fill the space

PointToPointHelper p2p;
p2p.SetDeviceAttribute("DataRate", StringValue("100Mbps")); // add the value
p2p.SetChannelAttribute("Delay", StringValue("5ms"));

NetDeviceContainer p2pDevices = p2p.Install(p2pNodes);

// Assign IP addresses
Ipv4AddressHelper address;

// fill the empty spaces for LAN1
address.SetBase("10.1.3.0", "255.255.255.0");
Ipv4InterfaceContainer lan3Interfaces = address.Assign(csmaNodes1);


// fill the empty spaces for LAN2
address.SetBase("10.1.2.0", "255.255.255.0 ");
Ipv4InterfaceContainer lan2Interfaces = address.Assign(csmaNodes);

// Fill the empty spaces for p2p
address.SetBase("10.1.1.0", " 255.255.255.0");
Ipv4InterfaceContainer p2pInterfaces = address.Assign(p2pDevices);

uint16_t echoPort = 93; //enter the number
UdpEchoServerHelper echoServer(echoPort);
ApplicationContainer serverApps = echoServer.Install(csmaNodes.Get (2)); //add the node
serverApps.Start(Seconds(1.0));
serverApps.Stop(Seconds(20.0));

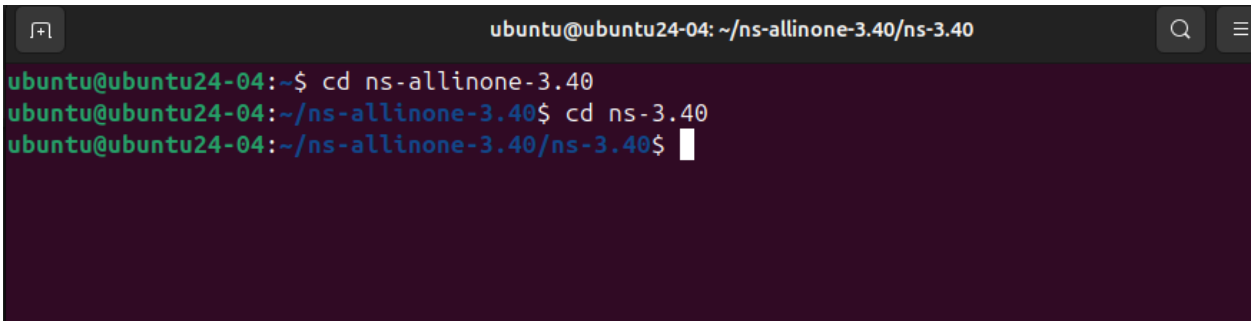
UdpEchoClientHelper echoClient( csmaInterfaces.GetAddress (2), echoPort); // add the node
echoClient.SetAttribute("MaxPackets", UintegerValue( 8)); //add the value
echoClient.SetAttribute("Interval", TimeValue(Seconds(1.0)));
echoClient.SetAttribute("PacketSize", UintegerValue(1024));

ApplicationContainer clientApps = echoClient.Install(csmaNodes1.Get(1)); //add the node
clientApps.Start(Seconds(2.0));
clientApps.Stop(Seconds(20.0));

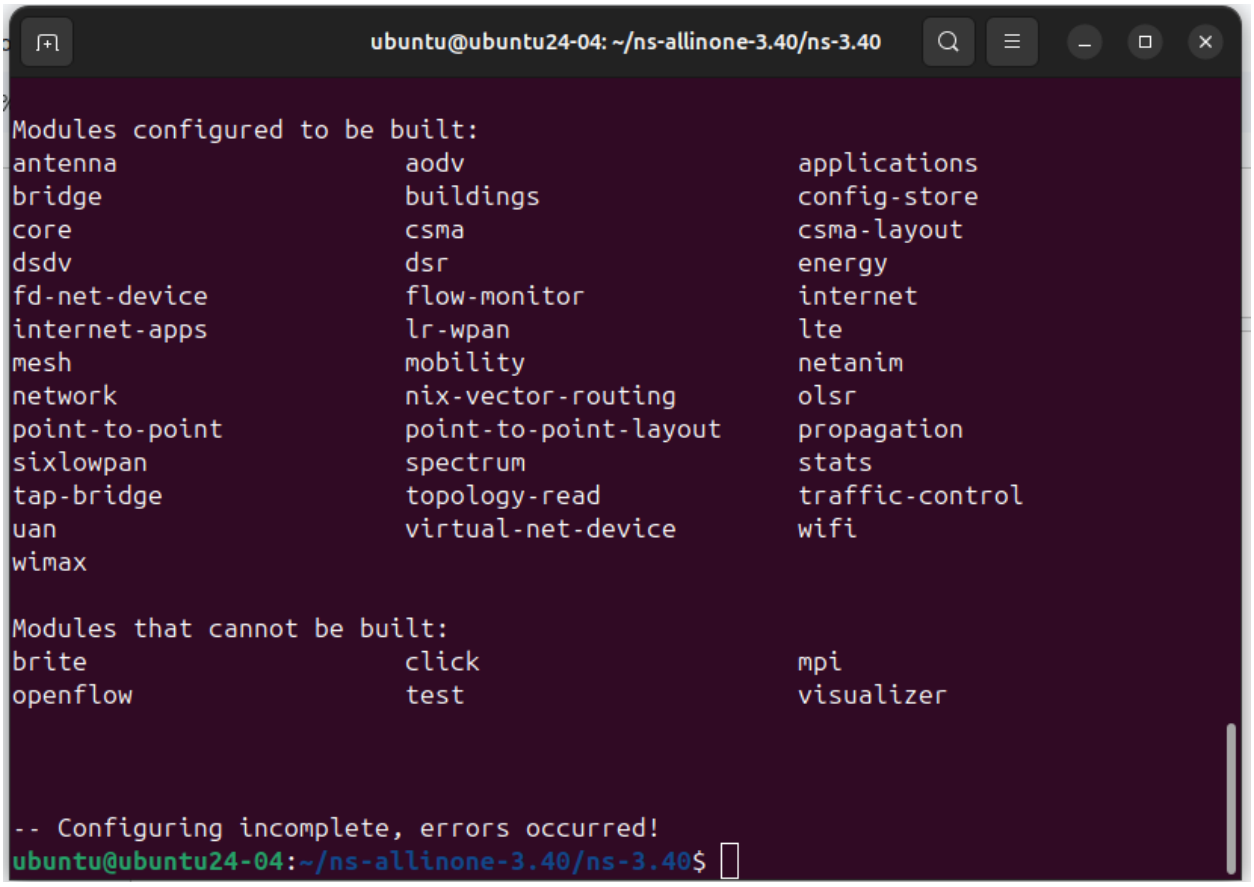
// Populate routing tables so nodes can reach each other
Ipv4GlobalRoutingHelper::PopulateRoutingTables();

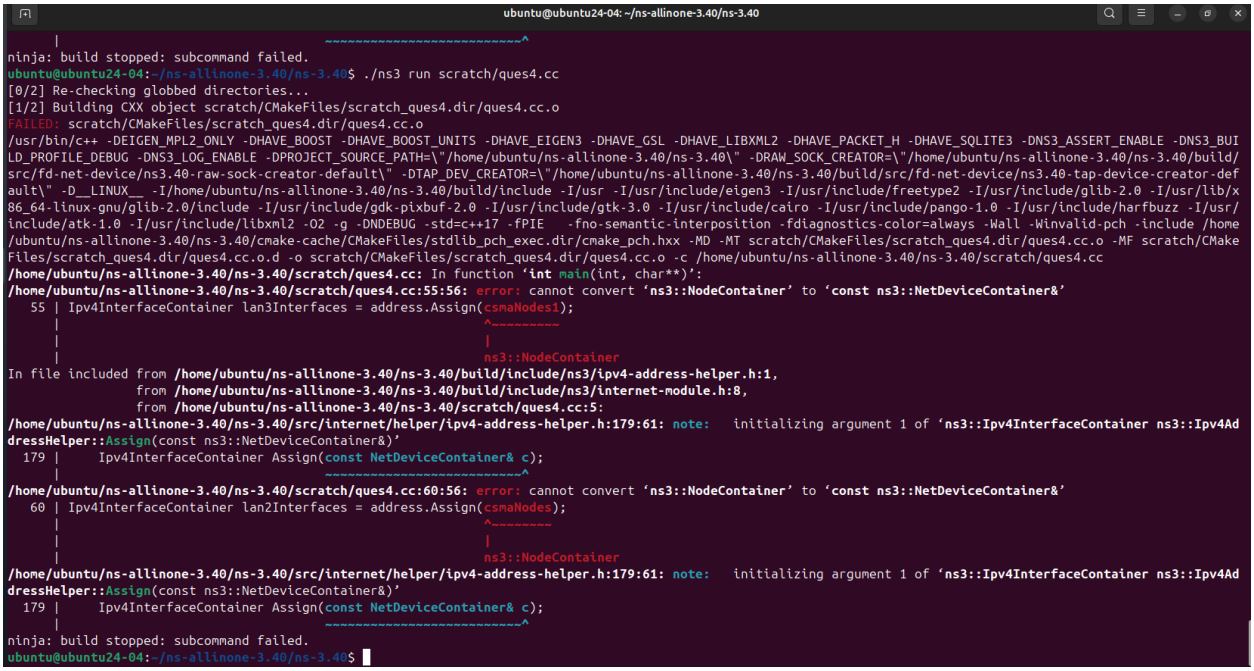
Simulator::Run();
Simulator::Destroy();
```

return 0;
}



Move cd scratch, Write nano to make file in scratch




ubuntu@ubuntu24-04: ~/ns-allinone-3.40/ns-3.40
ninja: build stopped: subcommand failed.
ubuntu@ubuntu24-04:~/ns-allinone-3.40/ns-3.40$ ./ns3 run scratch/ques4.cc
[0/2] Re-checking globbed directories...
[1/2] Building CXX object scratch/CMakeFiles/scratch_ques4.dir/ques4.cc.o
FAILED: scratch/CMakeFiles/scratch_ques4.dir/ques4.cc.o
/home/ubuntu/ns-allinone-3.40/ns-3.40/scratch/ques4.cc: In function 'int main(int, char**)':
/home/ubuntu/ns-allinone-3.40/ns-3.40/scratch/ques4.cc:55:56: error: cannot convert 'ns3::NodeContainer' to 'const ns3::NetDeviceContainer&'
ns3::NodeContainer
/home/ubuntu/ns-allinone-3.40/ns-3.40/scratch/ques4.cc:60:56: error: cannot convert 'ns3::NodeContainer' to 'const ns3::NetDeviceContainer&'
ns3::NodeContainer
ninja: build stopped: subcommand failed.
ubuntu@ubuntu24-04:~/ns-allinone-3.40/ns-3.40$